

DC-10s were judged prone to the same problem and temporarily grounded. The fatal vulnerability of Concorde's fuel tanks to a tire blow-out was revealed in the Paris crash in 2000.

Air-crash investigations, which can identify such faults, are obviously a vital part of the safety system. Few human activities are pursued quite as rigorously and painstakingly as the collection and examination of every fragment of a crashed aircraft. This sometimes occurs under the most trying circumstances. When a ValuJet DC-9 crashed in the alligator-infested Florida Everglades in May 1996, it was swallowed by the swamp leaving not a trace on the surface. Two months later, Boeing 747 Flight TWA 800 exploded in the air off Long Island, and investigators had to search the Atlantic seabed for pieces and reassemble the airliner like a giant jigsaw puzzle. But, finally, conclusions were reached in both these cases: the ValuJet DC-9 had been carrying inflammable cargo; an electrical fault on the TWA 747 had probably ignited fuel tanks.

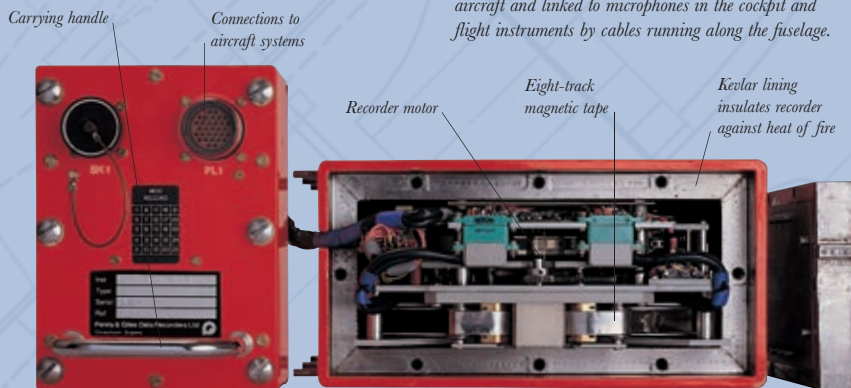
BLACK BOX RECORDERS

IN-FLIGHT RECORDERS have been mandatory on airliners since 1957. Although habitually referred to as black boxes, they are bright orange with reflective stripes. At first the boxes contained only a Flight Data Recorder (FDR), keeping a record of the aircraft's airspeed, altitude, compass headings, and so on. From 1966 a Cockpit Voice Recorder (CVR) was added. This has traditionally been a 30-minute magnetic tape loop – self-erasing, so that it always contains a record of the half hour before any crash. CVRs

are being upgraded to solid-state two-hour recorders. The black boxes are virtually indestructible. None has ever been crushed by force of impact, although a very few have been disabled by fire. If an aircraft crashes in the ocean, divers can still find the in-flight recorder by homing in on a locator beacon that emits an ultrasonic signal.

IN-FLIGHT RECORDER

The so-called black box is installed in the tail of the aircraft and linked to microphones in the cockpit and flight instruments by cables running along the fuselage.



RECONSTRUCTED FUSELAGE

This length of Boeing 747 fuselage was pieced together from fragments of TWA Flight 800, which crashed in the Atlantic shortly after takeoff from JFK Airport, New York, in August 1996. Thousands of pieces of the jet were retrieved from the ocean and painstakingly reassembled.

